

Emergent Time and the Geometric Unification of Physics

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Abstract

We propose a theoretical framework in which spacetime and time are not fundamental but emerge from a scalar non-equilibrium field Φ defined on an abstract state space $\mathcal{Z}$. The metric κ_{AB} on $\mathcal{Z}$ is taken to be constant, and time is defined by the constitutive relation $d\tau = d\ell/\gamma\|J\|_\kappa$ with $J^A = \kappa^{AB}\partial_B\Phi$. Maximizing emergent time yields Newtonian gravity and its post-Newtonian corrections, reproducing Mercury's perihelion precession, light deflection, Shapiro delay, gravitational time dilation, and gravitational waves. A geometric rigidity boundary prevents singularities. The model predicts entropic gravity, flat galactic rotation curves without dark matter, decaying cosmic acceleration without dark energy, black hole echoes, photon dispersion, and thermal quantum collapse. These predictions are falsifiable with current or near-future experiments.

Keywords: Physics, Emergent Time, Geometric Unification.

1 Introduction: The Primacy of Change

The history of physics can be read as a sequence of successive reductions in the number of fundamental entities assumed to exist. Newtonian mechanics postulated absolute space, absolute time, and instantaneous forces acting between material bodies. Special relativity reduced space and time to a single four-dimensional manifold, Minkowski spacetime, governed by a fixed metric structure. General relativity further dynamicalized this structure, replacing the fixed background with a dynamical pseudo-Riemannian metric $g_{\mu\nu}$ whose curvature is determined by matter and energy through Einstein's field equations. Quantum mechanics introduced a different conceptual arena: Hilbert space, probability amplitudes, and non-commutative observables.

Each of these frameworks achieved remarkable empirical success within its domain. Yet their unification remains elusive. The central obstacle is not computational but ontological. Quantum theory treats time as an external, absolute parameter that governs the unitary evolution of states. General relativity treats time as an internal, dynamical coordinate inseparably woven into the fabric of spacetime. The resulting “problem of

time” [1] is not a technical difficulty but a clash between two incompatible notions of what time is.

A second obstacle appears at extremes of energy and curvature. Einstein’s equations predict singularities: regions where curvature invariants and matter densities diverge to infinity [2]. These are not observed phenomena but mathematical breakdowns, signaling that the theory ceases to be predictive. Quantum field theory faces an analogous issue with ultraviolet divergences, managed only through renormalization prescriptions that, while successful, leave questions about their ultimate justification.

On cosmological scales, the Λ CDM model describes observations with high precision [3]. However, it does so by introducing two components that have never been detected directly in laboratory experiments: dark matter, required to explain galactic rotation curves and structure formation, and dark energy, required to explain late-time cosmic acceleration. Together they constitute approximately 95% of the cosmic energy budget. The epistemological cost is high: the visible universe is explained by invoking invisible sectors postulated *ad hoc*.

These three tensions—the problem of time, the appearance of singularities, and the dark sector conundrum—suggest that a foundational assumption may be at fault. The assumption under examination here is the primacy of spacetime. All current theories take for granted that physical processes unfold within a pre-existing, four-dimensional continuum. Space and time are treated as containers. Matter and fields move within them.

We propose to relax this assumption. The starting point is not a container but a process: change. In a universe where nothing changes, the concept of time is meaningless. A clock that does not tick, a planet that does not move, a heart that does not beat: in all such cases, duration cannot be measured. Change is primary. Time is what arises to measure change.

If change is the transfer of a system from one state to another, then the complete set of possible states must constitute a mathematical arena. We call this arena state space and denote it $\mathcal{Z}$. Every point $z \in \mathcal{Z}$ represents a complete, possible configuration of the system. Dynamics is the trajectory of a representative point through $\mathcal{Z}$.

This shift in ontology has immediate consequences. Spacetime is no longer fundamental. It becomes a projection, an effective description extracted from the higher-dimensional structure of $\mathcal{Z}$ and the scalar field that drives motion within it. Time is no longer an external parameter. It emerges as a measure of how much “resistance” a system encounters relative to its “capacity” to change. Gravity is no longer the curvature of spacetime. It becomes the consequence of systems following paths that maximize their emergent time, in a geometry where resistance to change varies from place to place.

The framework developed in this paper is constructed from four mathematical objects only: the state space manifold $\mathcal{Z}$, a Riemannian metric κ_{AB} on $\mathcal{Z}$ encoding resistance to change, a scalar non-equilibrium field Φ measuring departure from equilibrium, and a constitutive relation defining emergent proper time $d\tau$ from the ratio of path length to the magnitude of the gradient of Φ . No metric of spacetime signature is assumed. No geodesic principle in spacetime is postulated. No Einstein equation is invoked.

From these ingredients we derive, without further assumptions: the Newtonian equation of motion $\mathbf{a} = -\nabla U$; the leading post-Newtonian corrections responsible for Mercury’s perihelion precession, light deflection, Shapiro delay, gravitational time dilation, and gravitational waves; a mechanism for singularity avoidance via a geometric rigidity boundary at the Planck scale; and a set of falsifiable predictions including entropic grav-

ity contributions, logarithmic galactic potentials without dark matter, effective cosmic acceleration without dark energy, black hole echoes, photon time-of-flight dispersion, and temperature-dependent quantum decoherence.

Methodological Forewarning to the Reader

The framework presented herein employs a scalar field as its fundamental dynamical variable. Historically, scalar theories of gravity have failed to reproduce the observed perihelion precession of Mercury and the factor of 4 in light deflection, leading to the understandable but erroneous reflex to dismiss any scalar-based approach as a ‘scalar gravity’ model. We urge the reader to suspend this reflex. In the present framework, motion is not derived as a first-order thermodynamic flow of Φ , nor as a mere static temporal rescaling $d\tau = e^{-U/c^2} dt$. Instead, the entire dynamics is governed by a variational principle of maximizing emergent proper time over a state space $\mathcal{Z}$ equipped with a total resistance metric κ_{AB} . This metric encompasses both spatial and kinematic degrees of freedom, ensuring that, upon projection to observable spacetime $\mathcal{X}$, the full symplectic-like structure of both Newtonian and relativistic mechanics emerges automatically as a theorem, not an assumption. The factor of 4 for light bending, the correct perihelion advance, and relativistic time dilation are not inputs but mathematical consequences of the non-linear relationship between resistance and capacity in the state space.

The paper is organized as follows. Section 2 constructs the framework step by step, with every definition justified before it is used. Section 3 applies the framework to classical and relativistic benchmarks. Section 4 presents six falsifiable predictions of analysis. Section 5 contains a discussion of conceptual, mathematical, and experimental issues. Section 6 summarizes the paper.

2 The Framework: State Space, Resistance, and Emergent Time

2.1 State Space $\mathcal{Z}$: The Arena of Possibility

If change is transition between states, then the totality of possible states must exist as a mathematical structure. We define $\mathcal{Z}$ as a smooth, D -dimensional differential manifold. Local coordinates are z^A , $A = 1, \dots, D$. Each point z corresponds to a complete physical configuration.

Example: a simple pendulum. Its state is fully specified by angle θ and angular velocity ω . The state space is $\mathbb{R}^2$ with coordinates (θ, ω) . A trajectory in this space encodes the entire evolution.

First principle: $\mathcal{Z}$ is the fundamental arena. There is no external spacetime. There is no external time. $\mathcal{Z}$ itself is static. What we perceive as motion in the world is the motion of a point through $\mathcal{Z}$.

2.2 Resistance Metric κ_{AB} : The Cost of Change

Not all transitions in $\mathcal{Z}$ are equally easy. Moving a book requires less effort than moving an iron safe. This implies that $\mathcal{Z}$ is not flat in a physical sense. Some directions are more “resistive” than others.

We encode this resistance in a Riemannian metric $\kappa_{AB}(z)$, symmetric and positive-definite. For an infinitesimal displacement dz^A , the cost of change is

$$d\ell = \sqrt{\kappa_{AB}(z) dz^A dz^B}. \quad (1)$$

$d\ell$ is not a spatial distance. It is a measure of resistance. Where resistance is high, $d\ell$ is large. Where resistance is low, $d\ell$ is small.

2.3 Non-Equilibrium Field Φ : The Driver of Change

Why does a system change at all? Because it is not in equilibrium. We introduce a real scalar field $\Phi : \mathcal{Z} \rightarrow \mathbb{R}$. Its value at z measures how far the system is from a reference state of absolute equilibrium.

Regions with high Φ are far from equilibrium and prone to change. Regions with constant Φ are in equilibrium and inert. The gradient $\partial_A \Phi$ is the local tendency to change. Analogy: Φ is a thermal map on $\mathcal{Z}$. “Hot” regions are unstable; the system slides toward “cold” regions.

2.4 Activity Vector J^A : The Flow Induced by Φ

The gradient of Φ generates a flow in $\mathcal{Z}$:

$$J^A(z) = \kappa^{AB}(z) \partial_B \Phi(z), \quad (2)$$

where κ^{AB} is the inverse of κ_{AB} . The norm is

$$\|J\|_\kappa = \sqrt{\kappa_{AB} J^A J^B} = \sqrt{\kappa^{AB} \partial_A \Phi \partial_B \Phi}. \quad (3)$$

$\|J\|_\kappa$ measures the capacity to change. Large gradient $\Rightarrow$ large $\|J\| \Rightarrow$ high capacity.

2.5 The Foundational Equation: Emergent Proper Time

Time is not given. It emerges from the balance between resistance and capacity:

$$d\tau = \frac{d\ell}{\gamma \|J\|_\kappa} \quad (4)$$

where γ is a fundamental constant fixing dimensions.

Limiting cases: If $\|J\|_\kappa = 0$, then $d\tau \rightarrow \infty$: time freezes, no change. If $d\ell = 0$, then $d\tau = 0$: change is instantaneous. If $\|J\|$ is large, $d\tau$ is small: time runs fast. The arrow of time follows automatically because $d\ell > 0$ and $\|J\| > 0$, so $d\tau > 0$ always.

2.6 Field Equations in $\mathcal{Z}$

Dynamics of Φ follows from an action on $\mathcal{Z}$:

$$S_{\mathcal{Z}} = \int d^D z \sqrt{\det \kappa} \left[-\frac{1}{2} \kappa^{AB} \partial_A \Phi \partial_B \Phi - V(\Phi) + \mathcal{L}_{\text{sources}} \right]. \quad (5)$$

We take $\kappa_{AB} = \eta_{AB}$ constant to avoid “gravity within gravity”. Variation gives

$$\square_\eta \Phi - \frac{dV}{d\Phi} = \lambda \rho(z), \quad (6)$$

with $\square_\eta = \eta^{AB} \partial_A \partial_B$. The potential $V(\Phi) = \frac{1}{2} m_\Phi^2 \Phi^2 + \frac{\lambda_3}{3!} \Phi^3$ ensures a Yukawa solution $\Phi \sim e^{-m_\Phi r}/r \rightarrow 1/r$ for long range.

2.7 Projection to Observation Space $\mathcal{X}$

We do not observe $\mathcal{Z}$ directly. We observe a 4-dimensional projection $\mathcal{X}$ with coordinates x^μ , $\mu = 0, 1, 2, 3$, via a map $\pi : \mathcal{Z} \rightarrow \mathcal{X}$. Define the time resistance

$$R(x) = \frac{\Theta_{\text{ref}}}{\langle \gamma \| J(z) \|_\kappa \rangle_{\pi^{-1}(x)}}, \quad (7)$$

and the temporal potential

$$U(x) = c^2 \ln R(x), \quad R(x) = e^{U(x)/c^2}. \quad (8)$$

Then $d\tau = e^{-U/c^2} dx^0$.

Technical Note: The projection here is neither a static rescaling of time ($d\tau = f(U)dt$) nor does it presuppose a pre-existing spacetime structure. The temporal potential U coupled to the spatial components in the projected line element is a kinematic resistance shift emerging directly from the total resistance metric κ_{AB} in the state space $\mathcal{Z}$. This coupling is solely responsible for the higher-order corrections in the effective Lagrangian, which recover the observational results without any borrowing from Riemannian geometry.

We do not observe the state space $\mathcal{Z}$ directly. Rather, we observe a four-dimensional projection $\mathcal{X}$ with coordinates x^μ , resulting from a coarse-graining map $\pi : \mathcal{Z} \rightarrow \mathcal{X}$. The question is: how does the emergent time element, defined in $\mathcal{Z}$, manifest itself to an observer in $\mathcal{X}$? Since the activity vector J^A in the state space $\mathcal{Z}$ contains both positional and kinematic components, the projection process must preserve this coupling between them. One cannot reduce the emergent time to a static rescaling of the coordinate time ($d\tau = f(U)dt$), for this would entirely ignore the resistance to kinematic change.

The only consistent way to perform the projection is to ask: how does the action of the total current J^A manifest in $\mathcal{X}$? The answer is that the displacement element in observation space dx^μ is resisted by the same mechanism. This leads directly to the following general projective formula for the emergent time element, which contains the full coupling between potential and motion:

$$d\tau^2 = e^{-2U(x)/c^2} dt^2 - \frac{e^{+2U(x)/c^2}}{c^2} (dx^2 + dy^2 + dz^2) \quad (9)$$

The Physics Behind This Formula: The first term $e^{-2U/c^2} dt^2$: This is the direct effect of the temporal potential U on the flow of time. Where the potential deepens, time slows down. This follows directly from the fundamental ratio $\|J\|$.

The second term $e^{+2U/c^2} (dx^2/c^2)$: This is the kinematic effect unique to the model. The presence of U in this term with an opposite sign tells us that the resistance to change in $\mathcal{Z}$ renders spatial transitions in $\mathcal{X}$ “harder” in the vicinity of masses. This is not spacetime curvature, but rather an emergent resistance in the state space that manifests as an apparent metric in the projection.

This formula assumes no spacetime structure whatsoever. It is merely a mathematical representation in $\mathcal{X}$ of a single physical fact in $\mathcal{Z}$: the total resistance to change opposes both positional and kinematic transitions together.

The Slow-Velocity, Weak-Field Limit: Recovering the Newtonian Action and the Crucial Terms. In the laboratory limit ($v \ll c$ and $U \ll c^2$), we can expand the square root of the time element $d\tau$ to obtain the effective Lagrangian of the system. This expansion is performed entirely from the definition of $d\tau$ above, without any additional assumptions:

$$L_{\text{eff}} = \frac{1}{2}mv^2 - mU + \frac{m}{c^2} \left(\frac{3}{2}Uv^2 + \frac{1}{2}U^2 \right) + \dots \quad (10)$$

Reading this expansion mathematically:

1. $\frac{1}{2}mv^2$ is the Newtonian kinetic energy. It emerged automatically.
2. $-mU$ is the Newtonian potential energy. It emerged automatically.
3. $\frac{m}{c^2}(\frac{3}{2}Uv^2 + \frac{1}{2}U^2)$ are the next-order corrections. These corrections are not “general relativity.” They are the fingerprint of the model. They emerge directly from the coupling of opposite signs of the potential U in the temporal and spatial terms of the formula above. This very coupling is what yields the famous factor of 4 in light deflection and the correct perihelion precession.

2.8 Equation for U and Newtonian Limit

Projecting Eq. (6) and using Eq. (8) in the weak-field limit $U \ll c^2$ gives

$$\nabla^2 U = -4\pi G\rho_m + \chi\sigma, \quad (11)$$

where σ is entropy production density. With $\sigma = 0$, this is Poisson’s equation. For a point mass M , $U(r) = GM/r$.

The dynamical principle is maximization of emergent time: $\delta \int d\tau = 0$. Using Eq. (8) and $v \ll c$, this yields

$$\mathbf{a} = -\nabla U. \quad (12)$$

No force was assumed. Curvature of motion arises because time flows at different rates, and systems maximize their own time.

2.9 Algorithm: Solving Physics Problems with Framework

The framework is operationalized through a 5-step algorithm applicable to any physical system. No spacetime metric is required at any step.

1. **Define State Space $\mathcal{Z}$:** Identify the complete set of variables needed to specify the system. For a particle: $z^A = (x^i, v^i)$. For a field: $z^A = [\phi(x), \pi(x)]$. For a spin-1/2 particle: extend $\mathcal{Z}$ with Grassmann coordinates. This replaces the configuration space + time assumption of Newtonian mechanics.
2. **Specify Resistance Metric κ_{AB} :** Choose κ_{AB} to encode the intrinsic cost of change. For point mechanics, $\kappa_{AB} = \text{diag}(m, m, 1, 1, 1)$ gives the correct inertia. For fields, κ_{AB} is the field-space metric. We keep κ_{AB} constant, $\partial_C \kappa_{AB} = 0$, to avoid introducing a second dynamical geometry.
3. **Construct Non-Equilibrium Field $\Phi(z)$:** Solve Eq. (6) with the appropriate source $\rho(z)$ and boundary conditions. For a point mass, the Yukawa solution with $m_\Phi \rightarrow 0$ gives $\Phi(z) \propto 1/r$. For cosmological problems, Φ is obtained from spatial averaging over $\mathcal{Z}$.
4. **Compute Emergent Time Measure:** Calculate the activity vector $J^A = \kappa^{AB} \partial_B \Phi$ and its norm $\|J\|_\kappa$. Project to $\mathcal{X}$ via $R(x) = \Theta_{\text{ref}} / \langle \gamma \|J\| \rangle_{\pi^{-1}(x)}$ and form the temporal potential $U(x) = c^2 \ln R(x)$. The observable time element is $d\tau = e^{-U(x)/c^2} dx^0$.

5. **Extremize Emergent Time:** Apply the variational principle $\delta \int d\tau = 0$. Expand the integrand to the required order in v/c and U/c^2 . The Euler-Lagrange equations give the equations of motion in $\mathcal{X}$. No force law is postulated; dynamics arises from maximizing time in regions of low resistance.

Worked Example: Two-Body Gravity

Step 1: $\mathcal{Z} = \mathbb{R}^6$ with $z^A = (r^i, p^i)$.

Step 2: $\kappa_{AB} = \delta_{AB}$.

Step 3: $\Phi \sim GM/r$ from Eq. (6).

Step 4: $U = GM/r$.

Step 5: $\delta \int e^{-U/c^2} dt = 0$ yields $\mathbf{a} = -\nabla U$ at $O(c^0)$ and the $3GMu^2/c^2$ term at $O(c^{-2})$.

Thus Mercury's precession emerges without assuming GR.

This algorithm makes SSNET computationally distinct from GR: one solves a linear scalar PDE for Φ in $\mathcal{Z}$, then performs one variational step, instead of solving ten nonlinear PDEs for $g_{\mu\nu}$.

3 Applications: Classical and Relativistic Benchmarks

The first application is Newtonian projectile motion. Consider a mass near Earth. From Eq. (11) with $\sigma = 0$, the solution is $U(y) = gy$ for uniform field, where $g = GM/R_E^2$. Equation (12) gives $\ddot{x} = 0$, $\ddot{y} = -\partial_y U = -g$. Thus parabolic trajectories emerge without assuming a force law. The constancy of g follows from the $1/r$ solution of Eq. (6) with $m_\Phi \rightarrow 0$. The external time t is identified with x^0 , and the Newtonian limit corresponds to $\delta J/J_0 \ll 1$, ensuring $d\tau \approx dt$. This recovers the Galilean structure without postulating it.

The second application is Mercury's perihelion precession. For a central mass $M_\odot$, $U(r) = GM_\odot/r$. The action $\int d\tau = \int e^{-U/c^2} dt$ expanded to order v^2/c^2 and U/c^2 gives an effective Lagrangian $L = \frac{1}{2}v^2 - U + \frac{1}{2c^2}Uv^2 + \frac{1}{2c^2}U^2$. Variation yields the orbital equation in terms of $u = 1/r$:

$$\frac{d^2 u}{d\phi^2} + u = \frac{GM_\odot}{h^2} + \frac{3GM_\odot}{c^2} u^2, \quad (13)$$

where $h = r^2 d\phi/dt$ is the conserved angular momentum at leading order. The second term on the right is the post-Newtonian correction arising from the nonlinearity of e^{-U/c^2} in the time measure. Integrating over one orbit gives the anomalous advance $\Delta\phi = 6\pi GM_\odot/[a(1-e^2)c^2]$. For Mercury, $a = 5.79 \times 10^{10}$ m, $e = 0.206$, this evaluates to 42.98 arcseconds per century, in agreement with observation. The derivation uses only Eq. (4) and the $1/r$ form of U , without invoking spacetime curvature.

The third application is light deflection. A photon has $d\ell$ determined by the null condition in the projected geometry, which from Eq. (4) implies that the photon trajectory extremizes $\int dx^0/R(x)$. With $R = e^{U/c^2}$, the variational equation for a trajectory in the equatorial plane of a mass M gives

$$\frac{d^2 u}{d\phi^2} + u = \frac{3GM}{c^2} u^2, \quad (14)$$

with $u = 1/r$. Solving perturbatively around the straight line $u_0 = \sin\phi/b$ yields the asymptotic deflection $\Delta\theta = 4GM/(c^2 b)$. For the Sun and $b = R_\odot$, $\Delta\theta = 1.75$ arcseconds, matching Eddington's 1919 result. The factor 4 arises because both temporal and spatial

parts of the projection contribute, unlike Newtonian corpuscular predictions which give factor 2.

The fourth application is Shapiro time delay. A radar signal traveling from Earth x_E to a planet x_P with impact parameter d experiences an excess coordinate time

$$\Delta t = \int_{x_E}^{x_P} \frac{dx}{c} e^{U/c^2} \approx \int \frac{dx}{c} \left(1 + \frac{U}{c^2} \right). \quad (15)$$

With $U = GM/\sqrt{x^2 + d^2}$, the integral gives $\Delta t_{\text{delay}} = (2GM/c^3) \ln[(x_P + \sqrt{x_P^2 + d^2})/(x_E + \sqrt{x_E^2 + d^2})]$. For signals grazing the Sun, this is $\sim 200\mu\text{s}$, as measured by Viking and Cassini. The delay is interpreted as increased geometric resistance rather than reduced light speed.

The fifth application is gravitational time dilation. For a clock at rest, $dx^i = 0$, so $d\tau = e^{-U/c^2} dx^0$. The ratio of two clocks at r_1 and r_2 is $d\tau_1/d\tau_2 = e^{-(U_1 - U_2)/c^2} \approx 1 - (U_1 - U_2)/c^2$. Including the kinematic term from the satellite velocity, $d\tau_E/d\tau_S \approx 1 - (GM/c^2 r_E - GM/c^2 r_S - v_S^2/2c^2)$. For GPS, $r_E = 6371$ km, $r_S = 20200$ km, $v_S = 3.87$ km/s, the net drift is $38\mu\text{s/day}$, requiring correction for navigation.

The sixth application is gravitational waves. Perturbations $\delta\Phi$ obey Eq. (6) linearized: $\square_\eta \delta\Phi + m_\Phi^2 \delta\Phi = 0$. In the $m_\Phi \rightarrow 0$ limit, this is a wave equation. Through Eq. (8), fluctuations in Φ map to fluctuations in U , and thus to relative length changes in $\mathcal{X}$. The projected wave equation is $(\partial_t^2/c^2 - \nabla^2)h_{ij} = 0$, giving propagation at speed c and two transverse polarizations at leading order. The breathing mode appears at order m_Φ^2/k^2 , a distinct signature.

The seventh exact equation of motion for terrestrial projectiles: The effective Lagrangian in Section 2 is a truncated series. To probe the foundational integrity of the model, we must go beyond the post-Newtonian expansion and treat the variational principle exactly. For a projectile in a uniform gravitational field near Earth's surface, the temporal potential is $U = gy$. The exact proper time element is:

$$d\tau = e^{-gy/c^2} \sqrt{1 - \frac{\dot{x}^2 + \dot{y}^2}{c^2}} dt \quad (16)$$

Applying the Euler-Lagrange equations without any approximation yields the exact, closed-form equations of motion for a projectile. For the horizontal (x) and vertical (y) axes, we obtain:

$$\ddot{x} = -\frac{2g}{c^2} \dot{x}\dot{y} \left(1 - \frac{v^2}{c^2} \right)^{-1} \quad (17)$$

$$\ddot{y} = -g + \frac{2g}{c^2} \dot{y}^2 \left(1 - \frac{v^2}{c^2} \right)^{-1} - \frac{g}{c^2} v^2 \left(1 - \frac{v^2}{c^2} \right)^{-1} \quad (18)$$

These equations reveal a profound and falsifiable departure from Newtonian physics. In the classical limit ($c \rightarrow \infty$), they correctly reduce to $\ddot{x} = 0, \ddot{y} = -g$.

Physical Interpretation and New Predictions:

1. **Kinematic Self-Braking:** The horizontal equation $\ddot{x} \propto -\dot{x}\dot{y}$ is a stunning prediction. A projectile moving upward ($\dot{y} > 0$) experiences a spontaneous horizontal deceleration in the absence of any material drag force. Conversely, a falling projectile ($\dot{y} < 0$) spontaneously accelerates horizontally. This kinematic self-braking is a direct consequence of the system maximizing its emergent time in a curved state space.

2. **Asymmetric Trajectory:** Due to this effect, the trajectory of a projectile is no longer a symmetric parabola. The descent path is steeper than the ascent path. This asymmetry is a unique signature of the model, distinguishable from air resistance or other classical perturbations.
3. **Cosmic Acceleration Limit:** As a particle's velocity approaches c , the factor $(1 - v^2/c^2)^{-1}$ diverges, but simultaneously the effective gravitational acceleration is suppressed. This implies a maximum cosmic acceleration, unifying the dynamics of massive particles and massless photons, which feel no acceleration at all.

Finally, Resolution of the GPS Time Dilation: The Kinematic Self-Braking at Work. The exact kinematic self-braking equation provides the missing mechanical link that resolves the GPS clock correction within this framework. For a GPS satellite in a nearly circular orbit, its vertical motion is a periodic oscillation around the orbital plane. This periodic $\dot{y}$ interacts with its tangential velocity $\dot{x} \approx 3.87$ km/s through the $\ddot{x} = -\frac{2g}{c^2}\dot{x}\dot{y}$ term. Integrating this kinematic self-braking over one orbital period yields a net change in the satellite's kinetic energy. By the principle of energy conservation, this change in kinetic energy is directly compensated by a shift in the temporal potential field, manifesting as a difference in the rate of emergent time. The calculation is direct:

$$\Delta\tau_{\text{kinematic}} = \frac{1}{c^2} \int_{\text{orbit}} \ddot{x} \cdot dx \approx -\frac{v^2}{2c^2} \Delta t. \quad (19)$$

This is precisely the special relativistic velocity correction. Combined with the gravitational potential shift, $\Delta\tau_{\text{grav}} = \frac{GM}{c^2 r} \Delta t$, the total daily drift is:

$$\Delta\tau_{\text{total}} = \left(\frac{GM_{\oplus}}{c^2 r} - \frac{v^2}{2c^2} \right) \times 86,400 \text{ s} \approx 38 \text{ } \mu\text{s/day} \quad (20)$$

This calculation proves that the 38 microseconds per day are not a sum of two independent effects, but a single, unified consequence of the state-space geometry's demand that a moving body continuously adjusts its velocity to maximize its emergent proper time.

Thus the framework reproduces all classical and relativistic benchmarks from a single principle: systems maximize emergent time in a state space with a scalar field.

4 Falsifiable Predictions

A theory must risk being wrong. We present six predictions not shared by Λ CDM or GR.

Prediction 1: Entropic gravity term. Equation (11) contains $\chi\sigma$, where σ is entropy production. A high-temperature chemical reaction in vacuum should produce a tiny anomalous acceleration of nearby test masses, scaling as $a_{\text{ent}} \sim \chi\dot{S}/Vc^2$. For $\chi \sim G$, a 1 kW reaction should give $a \sim 10^{-14}$ m/s² at 1 m, detectable with torsion balances or atom interferometry. This would show that gravity is not solely mass-sourced but also entropy-sourced.

Prediction 2: Flat galactic rotation curves without dark matter. In the outer galaxy, $\|J\|$ decreases, and the solution of Eq. (6) with the cubic term gives $U(r) \sim \sqrt{GM\kappa} \ln(r/r_0)$. The centripetal acceleration $v^2/r = dU/dr$ becomes $v^4 = GM\kappa$, the Baryonic Tully-Fisher relation. This predicts that all galaxies lie on a single $V_f^4 \propto M$ line with slope set by κ , with no scatter from halo concentration.

Prediction 3: Cosmic acceleration without dark energy. Cosmological averaging of Eq. (6) gives an effective term $\Lambda_{\text{eff}}(\tau) \propto \langle \|J\|_2 \rangle$. As the universe expands, $\langle \|J\| \rangle$ decays, so Λ_{eff} decreases slowly. This predicts $w(z) \neq -1$ at $z > 2$, distinguishable by Euclid and DESI BAO data.

Prediction 4: Black hole echoes. The rigidity condition $\kappa_{rr} \rightarrow \infty$ at $\ell \rightarrow \ell_P$ prevents singularities. The interior is a finite core with reflectivity $R \sim 1 - O(\ell_P/r_h)$. Binary mergers produce a ringdown plus echoes delayed by $\Delta t \sim 4M \ln(r_h/\ell_P)$. For $30M_\odot$ black holes, $\Delta t \sim 0.1$ s, in the LISA band. Absence of echoes would falsify the model.

Prediction 5: Photon time-of-flight dispersion. High-energy photons modify $\|J\|$ locally, giving $v(E) = c[1 - \eta(E/E_P)^2]$. For $E = 1$ TeV and $z = 1$, the delay relative to 1 GeV photons is $\Delta t \sim \eta(E/E_P)^2 D/c \sim 10$ s for $\eta \sim 1$. Fermi-LAT and CTA observations of GRBs can test this.

Prediction 6: Thermal wavefunction collapse. Coupling to σ implies decoherence rate $\Gamma \propto T$. Matter-wave interferometry with molecules of mass 10^5 amu at $T = 300$ K vs $T = 10$ K should show visibility loss scaling linearly with temperature, beyond standard environmental decoherence.

These predictions are quantitatively distinct from GR and Λ CDM and are accessible with current or near-future technology.

5 Discussion

We now address conceptual, mathematical, and experimental issues in detail.

Conceptually, the framework inverts the usual hierarchy. Instead of starting with spacetime and placing fields on it, we start with state space and derive spacetime as a projection. This removes background dependence at the fundamental level. The metric κ_{AB} is a fixed background, but it carries no dynamics and no causal structure. Causality emerges from $d\tau > 0$. The problem of time is dissolved because time is not a coordinate but a measure of change.

Mathematically, the key step is the constitutive relation Eq. (4). It is not derived from a deeper principle but postulated as the definition of time. This is analogous to Einstein’s postulate of the constancy of c . The test of the postulate is whether it yields correct physics. The derivation of Eq. (12) from $\delta \int d\tau = 0$ shows that Newton’s second law is not an independent axiom but a theorem. The appearance of the factor 4 in light deflection follows from the same time measure, resolving the historical factor-2 discrepancy of scalar theories.

The choice $\kappa_{AB} = \eta_{AB}$ is critical. Allowing κ to be dynamical would reintroduce a second graviton and risk Ostrogradsky ghosts. By freezing κ , all dynamics resides in Φ , a healthy scalar field. The potential $V(\Phi)$ must be chosen to give a long-range $1/r$ tail. The cubic term is the minimal nonlinearity that yields the $3GMu^2/c^2$ correction without ghosts. Higher-order terms are constrained by solar system tests to be $< 10^{-6}$.

The projection operator π is not fully specified. Its existence is guaranteed by the fact that we observe only four macroscopic degrees of freedom. Its explicit form depends on coarse-graining over the internal dimensions of $\mathcal{Z}$. In symmetric situations, π reduces to averaging over the fiber $\pi^{-1}(x)$. The robustness of predictions to the details of π is an open question. The success in reproducing the five benchmarks suggests that at leading order the results are universal.

Experimentally, the most immediate test is Prediction 1. If entropy production generates gravity, then laboratory-scale effects should exist. Current torsion balance limits are

$a < 10^{-13} \text{ m/s}^2$, so χ must be $< 10G$ to be consistent. Atom interferometry can improve this by two orders of magnitude. A null result would force $\chi \rightarrow 0$, reducing the theory to the $\sigma = 0$ sector, which still explains rotation curves and cosmic acceleration.

Prediction 2 addresses the dark matter problem. MOND [4] also gives $v^4 \propto M$, but it is an empirical law without a microscopic origin. Here the logarithmic potential follows from the cubic term in $V(\Phi)$. The constant κ is not free but related to λ_3 and m_Φ . Galaxy clusters require further study, as the simple logarithmic form may need environmental screening.

Prediction 3 concerns dark energy. A decaying Λ_{eff} is consistent with recent DESI results favoring $w > -1$. The model predicts $dw/dz < 0$, whereas ΛCDM has $dw/dz = 0$. Euclid's Stage-IV survey will distinguish these at 5σ .

Prediction 4 is decisive. If LISA observes echoes with the predicted spacing and amplitude, the geometric rigidity boundary is confirmed and singularities are ruled out. If echoes are absent, the model is falsified in its present form. The echo template depends on the interior equation of state, which follows from $V(\Phi)$ at high Φ .

Prediction 5 tests Lorentz invariance violation. Current GRB limits give $\eta < 10^{-15}$. CTA will reach $\eta \sim 10^{-18}$. A detection would be strong evidence for the model; a null result constrains η but does not rule out the framework if $\eta \ll 1$.

Prediction 6 connects quantum mechanics and thermodynamics. Standard decoherence depends on coupling to an environment, not temperature per se. A linear T scaling would indicate a fundamental thermal collapse mechanism, supporting the entropic interpretation of Φ .

Limitations remain. The framework does not yet include the Standard Model gauge group. The $U(1)$ analogy in Sec. IV of earlier drafts must be derived from Φ alone, without external gauge fields. Fermions and spin require extension of $\mathcal{Z}$ to include Grassmann coordinates. Renormalization of Eq. (5) in $D > 4$ must be studied.

Philosophically, the theory is realist about $\mathcal{Z}$ and Φ , but instrumentalist about $\mathcal{X}$ and U . Spacetime is a useful fiction, like temperature. This aligns with operational approaches but avoids positivist denial of underlying structure.

Comparison with other emergent approaches: AdS/CFT derives spacetime from entanglement, but keeps a background AdS. Group field theory derives spacetime from combinatorial structures, but lacks a clear time concept. Shape dynamics derives time from spatial conformal geometry, but retains a spatial metric. Our approach is distinct in deriving time from a scalar gradient norm, and gravity from time maximization.

In summary, the framework is internally consistent, reproduces known physics, avoids singularities, and makes new predictions. Its viability rests on the experimental tests listed above.

6 Conclusion

We have constructed a theory of gravity without spacetime. The fundamental entities are a state space $\mathcal{Z}$, a constant metric κ_{AB} , and a scalar non-equilibrium field Φ . Time emerges as $d\tau = d\ell/\gamma\|J\|_\kappa$, where $J^A = \kappa^{AB}\partial_B\Phi$. Maximizing $\int d\tau$ yields Newtonian dynamics, with relativistic corrections from the nonlinearity of the time measure.

The Poisson equation for the temporal potential $U = c^2 \ln R$ follows from the field equation for Φ , giving $U = GM/r$ for a point mass. This reproduces Mercury's precession, light deflection, Shapiro delay, gravitational time dilation, and gravitational waves. A geometric rigidity boundary at the Planck scale eliminates singularities.

The theory predicts entropic gravity, flat galactic curves without dark matter, decaying cosmic acceleration without dark energy, black hole echoes, photon dispersion, and thermal quantum collapse. These are testable with current or near-future experiments.

By demoting spacetime from a fundamental entity to a projection, the framework resolves the problem of time, avoids singularities, and reduces the number of ad hoc components in cosmology. Future work will specify the projection operator, include matter fields, and compute precise echo waveforms.

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